

NITDGP/BTECH/Reg/Odd/2025-26

NATIONAL INSTITUTE OF TECHNOLOGY DURGAPUR
Odd Semester End-Term Examination, 2025-26

Course Code: CEC301
 Course Name: Solid Mechanics

Full Marks: 60
 Time: 3 Hours

Instructions: Assume missing data suitably. Neat and clean diagrams are mandatory for the solutions. Notations have their usual meanings if not mentioned otherwise.

Question No.	Body of the Question	Marks	Mapped CO
	Section A Attempt any THREE questions in Section A Q1 is compulsory. Further, attempt any one question from Q2 & Q3 and another question from Q4 & Q5.		
1.	a) Draw schematic diagram of a simple supported beam, of length 4 m and cross section (c/s) 250x300, in front view. Beside it, draw the cross section also. Hence, clearly mark the neutral axis of the cross section. Also specify the neutral surface of the beam using the co-ordinate system used. Assume beam material to be isotropic.	4	CO2
	b) Show typical bending stress distribution and shear stress distribution across a beam section where the cross section is a i) circular section and ii) "I section".	3	
	c) Write down the governing equation for finding out displacement at any point of a typical beam. Show the line diagram of the beam along with the relevant co-ordinate system; explain each term also.	3	
2.	a) Draw the following beams as directed. Hence find out the support reactions and draw SF and BM diagrams for each: i) Cantilever beam of length with concentrated clockwise moment M_0 at the centre and upward Force P_0 at the free end. Given: $P_0 = M_0/4L$ ii) Overhang beam, of overall length 3m and supported length 2m, and is subjected to a uniformly distributed downward load of 20 kN/m on the overhang part. b) Distinguish between a beam, a plate and a block in terms of their geometries. Draw relevant diagrams also to explain.	3 3 4	CO1
3.	A cantilever beam of span 3 m. has a circular cross section. Determine the diameter of the section in mm, if a uniformly distributed transverse load of intensity 4kN/m, spread throughout the beam, produces maximum shear stress of $8/3\pi$ N/mm ² . Hence, find out the maximum bending stress along with its location on the beam.	10	CO1
4.	Let there be an overhang beam ABC where BC is the overhang part. AB=4 m., BC= 2m. Let a uniformly distributed downward load of 2kN/m is acting on span BC. EI is constant. Estimate the deflection of the beam at the free end as well as at the centre of span AB and rotation at the roller support. Use Mecauley's method. Take the left end of the beam as origin, x is positive from left to right and deflection (y) downward is positive.	10	CO3

Course Outcomes

- CO1: Development of skills for predicting structural behavior of solids under different loads.
- CO2: Knowledge of basics of analysis and design of structural components made of variety of materials.
- CO3: Developing the requisite skill that helps in studying the advanced courses.